



MSc Thesis for Health Data Science

A Bayesian Time-Series Framework for Comparing Hospital Pressure Across Irish HSE Health Regions

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Declaration

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Master of Science is entirely our own work and had not been taken from the work of others or Artificial Intelligence, save and to the extent that such work has been cited and acknowledged within the text of our work.

ID: 25252089

Date:

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1 Introduction

2 Data

2.1 Data Collection

3 Methods

3.1 Data Collection Pipeline



Figure 1: Data pipeline from source to model input.

3.2 Model Development

Models were developed iteratively with increasing complexity as described in Gelman (2004). Model iteration can be visualised in appendix figure ??fig:model-inheritance). Model components were added based on residual patterns until the base model @eqref{eq:base-likelihood} was formed. First fitting the AR(1), annual cycle, and regional New Year. Then extending the model with 2 and 6 month cycles and an AR(2) component. Then tested against each other on which minimised the Deviance Information Criterion (DIC) and residuals. Table ?? reads as a record of this model selection. Below is a base model. The full model definitions can be found in Appendix 7.2.

The base model likelihood initialises at $t = 1$ and propagates an AR(1) correction for $t > 1$.

$$y_{i,1} \sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \quad (1)$$

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + \mathcal{E}_{i,t}^1, \tau_i^{-1}), \quad t > 1 \quad (2)$$

The AR(1) correction term captures week-to-week autocorrelation in the residuals.

$$\mathcal{E}_{i,t}^1 = \phi(y_{i,t-1} - \mu_{i,t-1}) \quad (3)$$

The mean function decomposes the expected trolley rate into a baseline level, an annual cycle, a regional New Year effect, and a Mid West funding reset.

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) + \mathcal{R}_{i,t} + \mathcal{W}_{i,t} \quad (4)$$

The regional New Year effect allows each region to respond differently across the three-week New Year window.

$$\begin{aligned} \mathcal{R}_{i,t} = & \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) \\ & + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) \\ & + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \end{aligned} \quad (5)$$

The Mid West funding reset captures a step-change in trolley pressure coinciding with the HSE Mid West restructuring across three weeks in 2021.

$$\begin{aligned} \mathcal{W}_{i,t} = & \mathbf{1}(i = \text{HSE Mid West}) \cdot (\sigma_{\text{pre}} \mathbf{1}(t = 86) \\ & + \sigma_{\text{mid}} \mathbf{1}(t = 87) \\ & + \sigma_{\text{post}} \mathbf{1}(t = 88)) \end{aligned} \quad (6)$$

3.3 Response Scaling

The on trolley counts were standardised to 10,000 people in each health region, Raw trolley counts vary across regions primarily because of population size rather than differences in hospital pressure. To enable fair comparison, counts were standardised to a rate per 10,000 catchment population.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region catchment population}_i} \times 10,000$$

This transforms the outcome into patients on trolley per 10,000 population per week, removing the confounding effect of region size and allowing direct comparison of hospital pressure across the six HSE health regions.

3.4 Posterior Sampling and Bayesian Ranking

Posterior sampling was performed using *Just another Gibbs sampler (JAGs)* (Plummer, 2003), implemented by pyJAGS (Nowotny, 2024), a Python interface to JAGS. Health region baselines, α_j^s , were used for ranking. Where j is one of the 6 HSE health regions, $j \in \{1, \dots, 6\}$, and s is the draw index, for $s \in \{1, \dots, n\}$ and total samples is defined as $n = 20,000$.

$$A = \begin{bmatrix} \alpha_1^1 & \dots & \alpha_j^1 & \dots & \alpha_6^1 \\ \vdots & & \vdots & & \vdots \\ \alpha_1^s & \dots & \alpha_j^s & \dots & \alpha_6^s \\ \vdots & & \vdots & & \vdots \\ \alpha_1^n & \dots & \alpha_j^n & \dots & \alpha_6^n \end{bmatrix}$$

Ranks follow the posterior ranking approach of Laird & Louis (1989). Where each sample is posterior sample is ranked to form rank distribution, R .

$$R = \begin{bmatrix} r_1^1 & \dots & r_j^1 & \dots & r_6^1 \\ \vdots & & \vdots & & \vdots \\ r_1^s & \dots & r_j^s & \dots & r_6^s \\ \vdots & & \vdots & & \vdots \\ r_1^n & \dots & r_j^n & \dots & r_6^n \end{bmatrix}$$

Where $r_j^s = \text{rank}(\alpha_j^s) = \sum_{i=1}^6 I(\alpha_j^s > \alpha_i^s)$. Given a tie, ranks are increased by $\frac{1}{2}$ steps for each tied value:

$$r_j^s = \sum_{i=1}^6 [\mathbf{1}(\alpha_j^s > \alpha_i^s) + \frac{1}{2}\mathbf{1}(\alpha_j^s = \alpha_i^s)]$$

Then posterior rank averages, $\hat{r}_i$, are calculated taking the mean across samples.

$$\hat{r}_j = \frac{1}{n} \sum_{s=1}^n r_j^s$$

$$\hat{R} = [\hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6]$$

Then posterior ranked averages are ranked once again to achieve ranked posterior ranked averages are calculated.

$$r_j^* = \text{rank}(\alpha_j^*) = \sum_{i=1}^6 I(\alpha_j^* > \alpha_i^*)$$

$$R^* = [r_1^*, r_2^*, r_3^*, r_4^*, r_5^*, r_6^*]$$

Where $r_j^* = \text{rank}(\hat{r}_j) = \sum_{i=1}^6 I(\hat{r}_j > \hat{r}_i)$.

3.5 Code and Data Availability

[Placeholder: link to GitHub repository, languages and tools used (Python, R, JAGS), and note that source data is publicly available from the HSE Urgent and Emergency Care portal.]

4 Results

4.1 Model Comparison

Model fit was compared using the DIC. A DIC table was generated at every response scale.

4.1.1 Scaled by 2022 health region population (per 10,000 people)

4.1.2 Regional Baseline

4.1.3 Week-to-Week Autocorrelation

4.1.4 Annual Cycle

4.1.5 Regional Rankings

5 Discussion

6 References

- Gelman, A. (2004). Exploratory Data Analysis for Complex Models. *Journal of Computational and Graphical Statistics*, 13(4), 755–779. <https://doi.org/10.1198/106186004X11435>
- Laird, N. M., & Louis, T. A. (1989). Empirical Bayes ranking methods. *Journal of Educational Statistics*, 14(1), 29–46. <https://doi.org/10.3102/10769986014001029>
- Nowotny, M. (2024). *pyJAGS: A Python interface to JAGS* (Version 2.4.0) [Computer software]. <https://github.com/michaelnowotny/pyjags>
- Plummer, M. (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*. <https://www.r-project.org/conferences/DSC-2003/Proceedings/Plummer.pdf>

7 Appendix

7.1 Model Inheritance

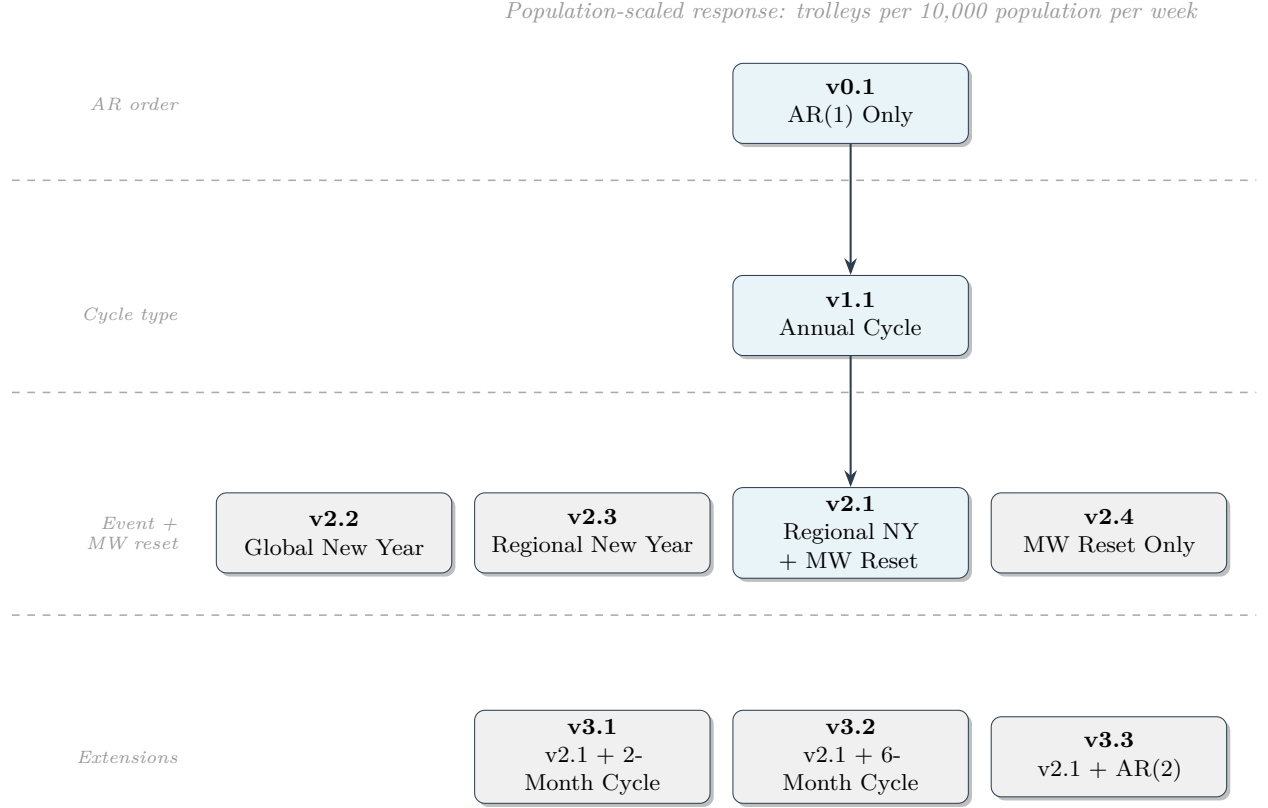


Figure 2: Model inheritance diagram. Spine nodes (blue) form the selected model sequence; grey nodes are tested variants. Thick arrows indicate the selected path; thin arrows indicate branches.

7.2 Full Model Specifications

Model v0.1

- AR(1) baseline; no cycle, no event effects

Likelihood

$$y_{i,1} \sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1})$$

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + \mathcal{E}_{i,t}^1, \tau_i^{-1}), \quad t > 1$$

Where $\mu_{i,t}$ is the baseline mean for region i at time t .

$$\mu_{i,t} = \alpha_i$$

$\mathcal{E}_{i,t}^1$ is the AR(1) error term.

$$\mathcal{E}_{i,t}^1 = \phi(y_{i,t-1} - \mu_{i,t-1})$$

Priors

$$\begin{aligned}\alpha_i &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v1.1

- Extends v0.1; adds annual cycle

Likelihood

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1\end{aligned}$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v2.1

- Extends v1.1; adds regional New Year effect and Mid West funding reset

Likelihood

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \\ &\quad + (\sigma_{\text{pre}} \mathbf{1}(t = 86) + \sigma_{\text{mid}} \mathbf{1}(t = 87) + \sigma_{\text{post}} \mathbf{1}(t = 88)) \cdot \text{mw}_i \\ y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1\end{aligned}$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\ \sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v2.2

- Extends v1.1; adds global New Year effect (shared across regions)

Likelihood

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + \delta_{\text{pre}} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid}} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post}} \mathbf{1}(t \bmod 52 = 2) \\ y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1\end{aligned}$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \delta_{\text{pre}}, \delta_{\text{mid}}, \delta_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v2.3

- Extends v1.1; adds regional New Year effect (no MW reset)

Likelihood

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \\ y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1\end{aligned}$$

Priors

$$\begin{aligned}\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\ \delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\ \tau_i &\sim \Gamma(0.001, 0.001) \\ \phi &\sim \text{Uniform}(-1, 1)\end{aligned}$$

Model v2.4

- Extends v1.1; adds Mid West funding reset (no New Year effect)

Likelihood

$$\begin{aligned}\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\ &\quad + (\sigma_{\text{pre}} \mathbf{1}(t = 86) + \sigma_{\text{mid}} \mathbf{1}(t = 87) + \sigma_{\text{post}} \mathbf{1}(t = 88)) \cdot \text{mw}_i \\ y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\ y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1\end{aligned}$$

Priors

$$\begin{aligned}
\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\
\sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\
\tau_i &\sim \Gamma(0.001, 0.001) \\
\phi &\sim \text{Uniform}(-1, 1)
\end{aligned}$$

Model v3.1

- Extends v2.1; adds 8-week sub-annual cycle

Likelihood

$$\begin{aligned}
\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\
&\quad + \beta_{2,i} \cos\left(\frac{2\pi t}{8}\right) + \gamma_{2,i} \sin\left(\frac{2\pi t}{8}\right) \\
&\quad + \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \\
&\quad + (\sigma_{\text{pre}} \mathbf{1}(t = 86) + \sigma_{\text{mid}} \mathbf{1}(t = 87) + \sigma_{\text{post}} \mathbf{1}(t = 88)) \cdot \text{mw}_i \\
y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\
y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1
\end{aligned}$$

Priors

$$\begin{aligned}
\alpha_i, \beta_i, \gamma_i, \beta_{2,i}, \gamma_{2,i} &\sim \mathcal{N}(0, 0.001) \\
\delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\
\sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\
\tau_i &\sim \Gamma(0.001, 0.001) \\
\phi &\sim \text{Uniform}(-1, 1)
\end{aligned}$$

Model v3.2

- Extends v2.1; adds 26-week sub-annual cycle

Likelihood

$$\begin{aligned}
\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\
&\quad + \beta_{2,i} \cos\left(\frac{2\pi t}{26}\right) + \gamma_{2,i} \sin\left(\frac{2\pi t}{26}\right) \\
&\quad + \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \\
&\quad + (\sigma_{\text{pre}} \mathbf{1}(t = 86) + \sigma_{\text{mid}} \mathbf{1}(t = 87) + \sigma_{\text{post}} \mathbf{1}(t = 88)) \cdot \text{mw}_i \\
y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\
y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1
\end{aligned}$$

Priors

$$\begin{aligned}
\alpha_i, \beta_i, \gamma_i, \beta_{2,i}, \gamma_{2,i} &\sim \mathcal{N}(0, 0.001) \\
\delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\
\sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\
\tau_i &\sim \Gamma(0.001, 0.001) \\
\phi &\sim \text{Uniform}(-1, 1)
\end{aligned}$$

Model v3.3

- Extends v2.1; replaces AR(1) with AR(2)

Likelihood

$$\begin{aligned}
\mu_{i,t} &= \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \\
&\quad + \delta_{\text{pre},i} \mathbf{1}(t \bmod 52 = 0) + \delta_{\text{mid},i} \mathbf{1}(t \bmod 52 = 1) + \delta_{\text{post},i} \mathbf{1}(t \bmod 52 = 2) \\
&\quad + (\sigma_{\text{pre}} \mathbf{1}(t = 86) + \sigma_{\text{mid}} \mathbf{1}(t = 87) + \sigma_{\text{post}} \mathbf{1}(t = 88)) \cdot \text{mw}_i \\
y_{i,1} &\sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \\
y_{i,2} &\sim \mathcal{N}(\mu_{i,2} + \phi_1 (y_{i,1} - \mu_{i,1}), \tau_i^{-1}) \\
y_{i,t} &\sim \mathcal{N}(\mu_{i,t} + \phi_1 (y_{i,t-1} - \mu_{i,t-1}) + \phi_2 (y_{i,t-2} - \mu_{i,t-2}), \tau_i^{-1}), \quad t > 2
\end{aligned}$$

Priors

$$\begin{aligned}
\alpha_i, \beta_i, \gamma_i &\sim \mathcal{N}(0, 0.001) \\
\delta_{\text{pre},i}, \delta_{\text{mid},i}, \delta_{\text{post},i} &\sim \mathcal{N}(0, 0.001) \\
\sigma_{\text{pre}}, \sigma_{\text{mid}}, \sigma_{\text{post}} &\sim \mathcal{N}(0, 0.001) \\
\tau_i &\sim \Gamma(0.001, 0.001) \\
\phi_1, \phi_2 &\sim \mathcal{N}(0, 0.01)
\end{aligned}$$

7.3 Alternative Response Scales

The primary analysis uses population-scaled trolley counts (per 10,000 catchment population). Two sensitivity scalings were also fitted to assess robustness of model selection and regional rankings to the choice of denominator.

7.3.1 Budget scaling

Weekly trolley counts scaled by the 2026 HSE regional budget allocation, expressing the outcome as trolleys per €1 billion allocated to the region.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region budget}_i / 10^6}$$

where health region budget_{*i*} is the 2026 budget allocation for region *i* in €thousands. This controls for funding differentials: a region receiving a larger allocation is expected to handle a proportionally larger patient volume.

7.3.2 Bed scaling

Weekly trolley counts scaled by the number of inpatient beds in each region, expressing the outcome as an overcapacity rate: for every 100 inpatient beds, how many patients were waiting on trolleys that week.

$$y_{i,t} = \frac{\text{weekly trolley count}_{i,t}}{\text{health region inpatient beds}_i} \times 100$$

where health region inpatient beds_{*i*} is the total inpatient bed count across hospitals in region *i*, summed from the June 2025 bed census. Hospitals with missing bed counts are excluded from the regional sum.

7.4 Residual Diagnostics

[Residual diagnostic plots for each model.]

7.5 Posterior Mean Fit

[Posterior mean $\mu_{i,t}$ overlaid on observed $y_{i,t}$ for each spine model.]

7.6 Gelman-Rubin Convergence Summary

[Gelman-Rubin $\hat{R}$ summary table for each spine model.]

7.7 Posterior Parameter Estimates

[Posterior means and 95% credible intervals for each spine model.]